SerpentGuard AI — Dataset Preparation, SAM 2.1 Masks & EDA



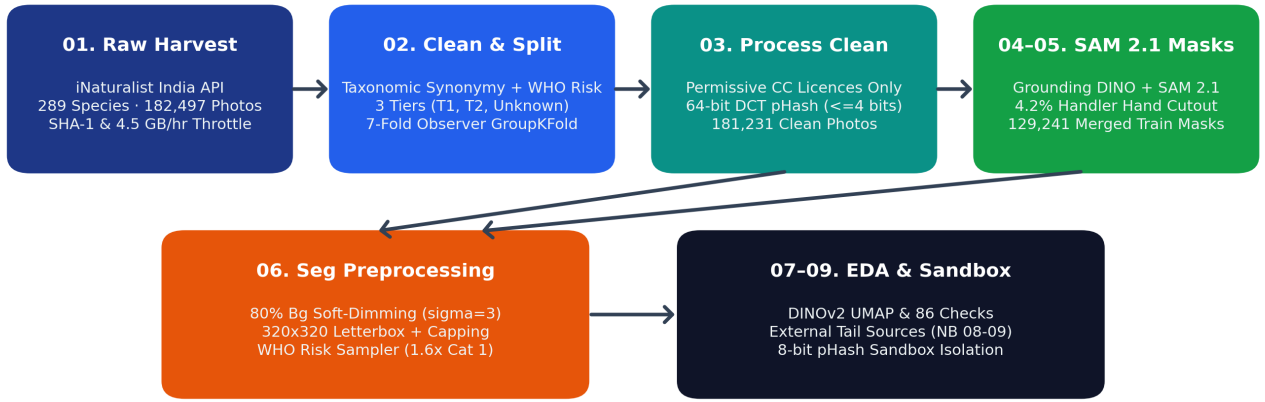
Offline Edge Vision for Indian Snake Encounters

- **Research Goal**: Cautious, offline Android decision support combining **SAM 2.1 foreground segmentation** and **hierarchical classification** across India's 289 snake species.
- **Reproducible 10-Notebook Suite**: End-to-end automated pipeline from iNaturalist API harvesting to leak-free splits, SAM 2.1 hand removal, and PyTorch safety DataLoaders.
- **Safety-First Abstention**: Rare species (<20 photos) are isolated in an open-set `unknown_pool`—never misclassified as harmless.

CLEAN DATASET	ACTIVE TAXONOMY	CLINICAL PRIORITY	SEGMENTATION MASKS
181,231	**208 Classes**	**18 WHO Species**	**129,241 Masks**
PERMISSIVE-LICENSED PHOTOS	68 TIER-1 · 140 TIER-2 (+81 UNKNOWN)	6 CAT 1 CRITICAL · 12 CAT 2 IMPORTANT	GROUNDING DINO + SAM 2.1 CONTOURS

Group 5: Jayesh Bansal (Team Lead) · Vikas · Puneet Bajaj · Manas Rastogi

Six-Stage Pipeline From Raw iNaturalist API to PyTorch Tensors

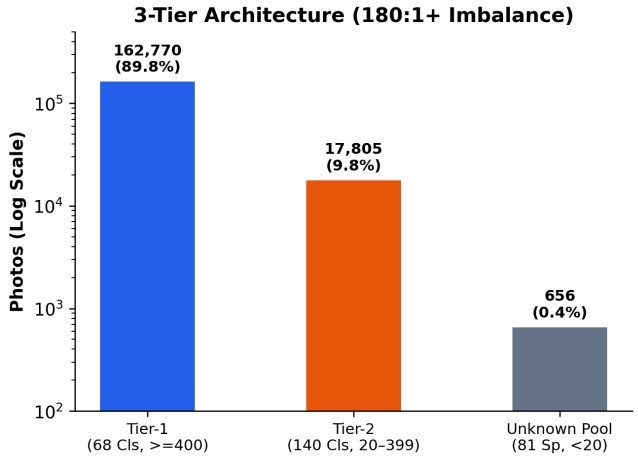


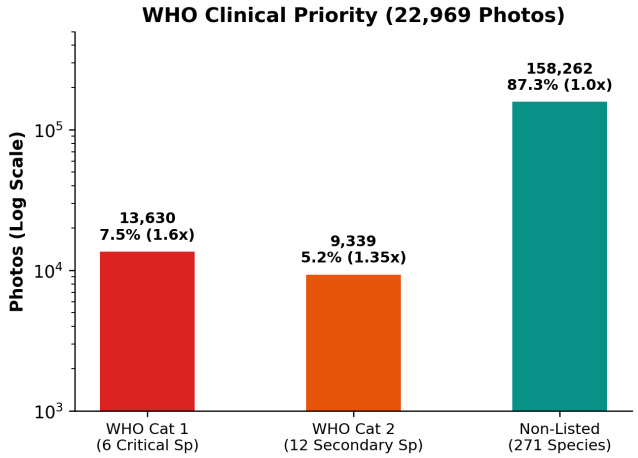
Automated & Auditable Flow

- **Harvest & Hygiene (NB 01–02)**: Queries iNaturalist (`taxon_id=85553`, throttled at 4.5 GB/hr), drops corrupt/tiny files (<5 KB or <100 px), and resolves taxonomic synonymy.
- **Clean Production Release (NB 03)**: Filters to permissive CC licences and purges cross-split near-duplicates via **64-bit DCT pHash (≤4 bits)**.
- **Parallel GPU Masking & Loader (NB 04–06)**: Merges sharded SAM 2.1 masks and feeds our 320×320 soft-dimmed PyTorch DataLoader.

PUBLISHED KAGGLE ARTEFACT	SOURCE NOTEBOOKS	CORE CONTENTS & ROLE
`serpentguard-m2-splits & snakes-clean`	Notebooks 01, 02, 03	181,231 deduplicated photos in `data/<split>/<species>/`, `metadata.csv`, `classes.csv`, and attribution grids
`serpentguard-sam-masks & extended`	Notebooks 04, 05, 08, 09	129,241 merged SAM 2.1 masks (`mask_stats.csv`) + isolated external tail-augmentation sandbox

289 Indian Species Organized into 3 Tiers & WHO Venom Risk Levels



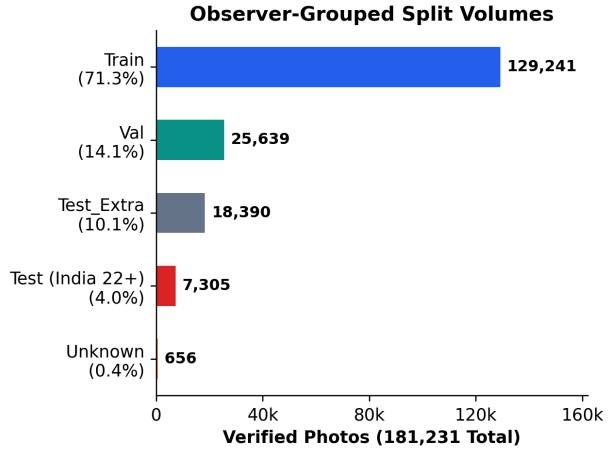


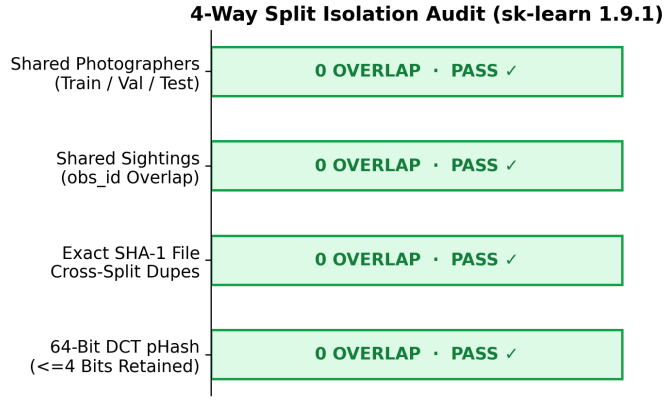
Solving the 180:1+ Long Tail

- **Three-Tier Partition**: **Tier-1** (≥400 photos: 68 classes, 89.8% of data), **Tier-2** (20–399 photos: 140 classes, 9.8%), and **Unknown Pool** (<20 photos: 81 species held out for open-set testing).
- **WHO Category 1 & 2 Priority**: Covers India's **Big Four** (*Naja naja, Daboia russelii, Bungarus caeruleus, Echis carinatus*) plus 12 secondary venomous species (22,969 photos).
- **Multi-Task Hierarchy**: Every image carries 4 labels: `species` (208), `genus` (77), `family` (14), and `who_risk` (3).

TIER / RISK GROUP	SPECIES / CLASSES	PHOTO VOLUME	TRAINING & EVALUATION STRATEGY
Tier-1 (≥400) & Tier-2 (20–399)	68 Tier-1 · 140 Tier-2 (208 Classes)	162,770 (89.8%) · 17,805 (9.8%)	Cap head classes at 1,000 photos; track macro-F1 separately across Tier-1 and Tier-2
Unknown Pool (<20) & WHO Cat 1/2	81 Unknown sp · 18 WHO sp	656 Unknown · 22,969 WHO (12.7%)	Calibrate open-set abstention on Unknown Pool; apply 1.6×/1.35× safety boost to WHO Cat 1/2

Photographer-Grouped Splits & 64-Bit DCT pHash Deduplication



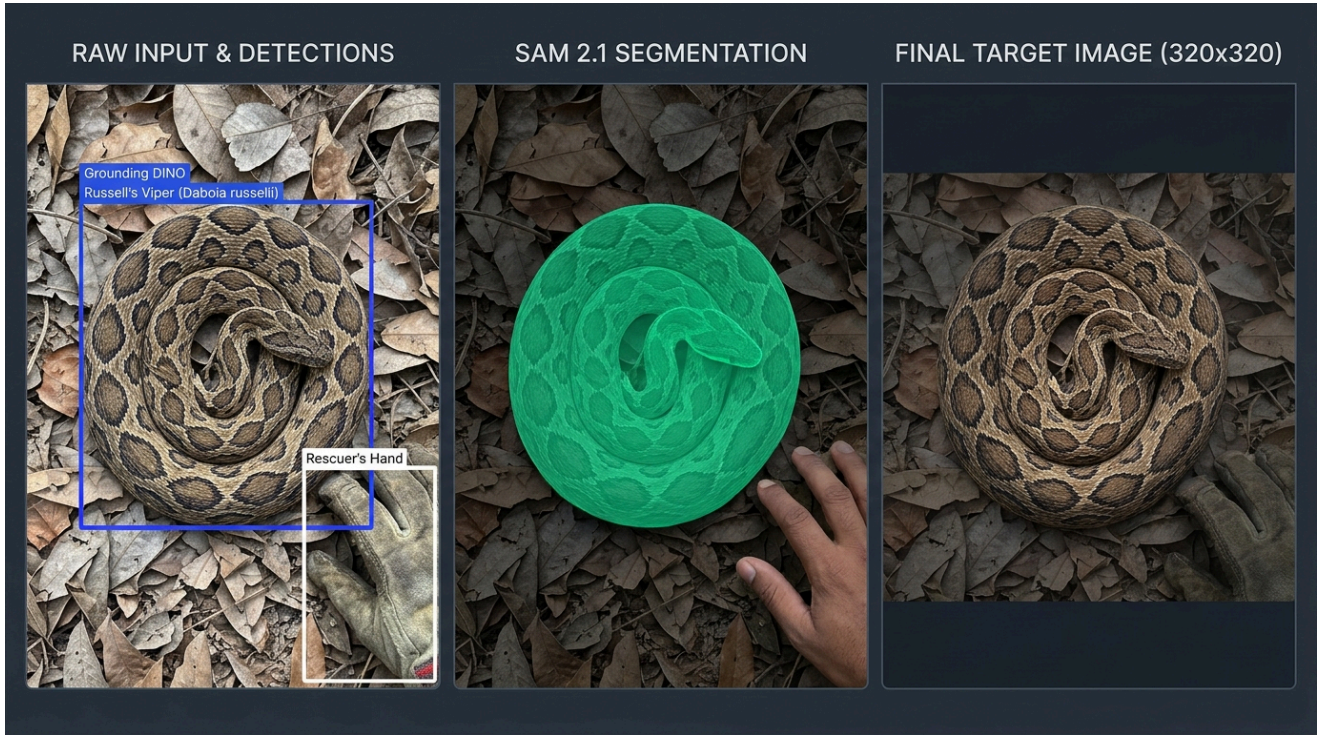


Why Grouped Splitting Is Mandatory

- **Zero Photographer Overlap**: `StratifiedGroupKFold` (7 folds grouped by `user_id`, `random_state=42`, pinned to `scikit-learn==1.9.1`) keeps all burst photos from a photographer in a single split.
- **Cross-Split pHash Purge**: 64-bit DCT perceptual hashing removes near-duplicate uploads (Hamming distance ≤4 bits) across splits.
- **Safe Loading Rule (NB 10)**: Always map targets via `classes.csv` rather than folder indices, since rare classes have 0 photos in `test`.

INTEGRITY GATE	METHOD ENFORCED IN NOTEBOOKS 02 & 03	VERIFIED OUTCOME
Observer & Sighting Isolation	Grouped assignment on photographer `observer_id` across `train`, `val`, and `test`	**0 shared photographers** and **0 shared sightings** across splits
Duplicate & Licence Filtering	SHA-1 exact dedup (-928) + cross-split pHash ≤4 bits (-272) + CC licence filter	**0 duplicate leakage**; 100% compliant attribution metadata

Grounding DINO + SAM 2.1 Masks with Automated Rescuer Hand Removal

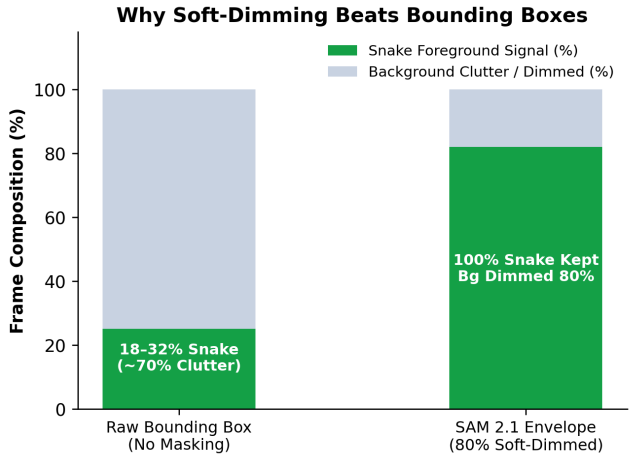


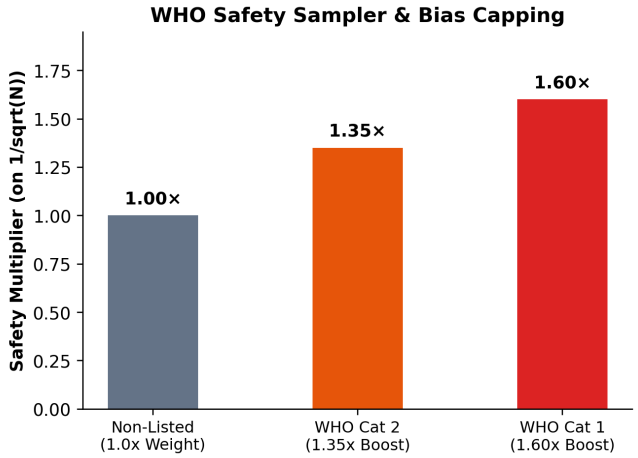
Eliminating Background & Handler Bias

- **Why Not Bounding Boxes?** Snake bodies occupy just **18%–32%** of the image area—raw bounding boxes retain up to **70% background clutter** (leaves, soil, concrete).
- **Two-Stage Mask Pipeline**: **Grounding DINO** (`grounding-dino-tiny`, prompt *"a snake."*) boxes feed into **SAM 2.1** (`sam2.1-hiera-large`) across parallel GPU shards.
- **Rescuer Hand Subtraction (~4.2%)**: Auxiliary hand/glove detector subtracts overlapping human hands so the classifier never learns hands as a snake feature.

SAM 2.1 PIPELINE STAGE	MODEL / QUALITY GATE	DATASET IMPACT ACROSS 129,241 TRAINING PHOTOS
Detection & Pixel Contouring	Grounding DINO + SAM 2.1 Hiera-Large	100% training split coverage; isolates exact dorsal scale contours from habitat clutter
Hand Cutout & Quality Flagging	Hand/glove subtraction + flags (<0.25 score or >95% area)	Removed handler hands in **~4.2% (5,843 photos)**; flags low-confidence masks for full-frame fallback

80% Background Soft-Dimming, 320×320 Letterboxing & WHO Safety Sampler



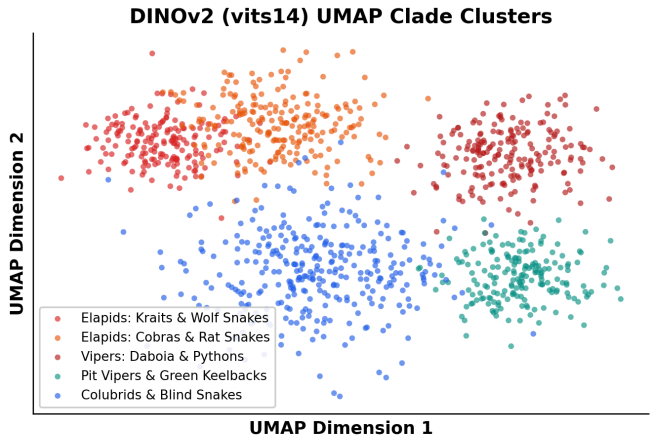


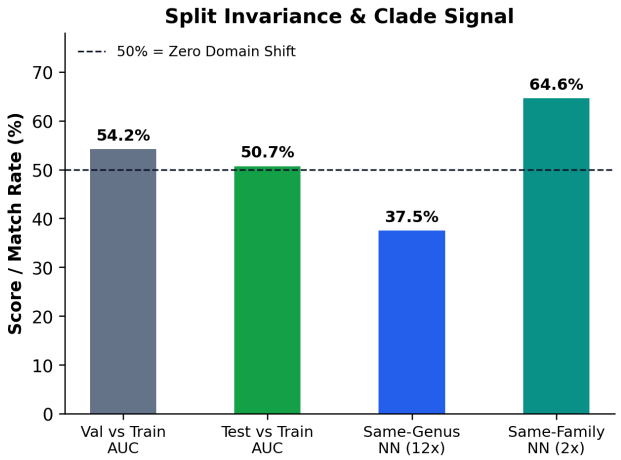
Bridging Masks to PyTorch Tensors

- **SAM Soft-Dimming (σ = 3)**: Retains **100% of snake pixels**, dims background by **80%** (`0.20 ×` bg), and smooths mask borders to prevent harsh cutout edges.
- **320×320 Letterbox (+6% Margin)**: Preserves natural body girth (slender vine snakes vs. stout vipers) and corrects underexposed night photos.
- **Bias Capping & Safety Sampler**: Caps classes at **1,000 photos** and photographers at **50 photos**, paired with a $1 / \sqrt{N_c}$ sampler boosted **1.6× for WHO Cat 1** and **1.35× for Cat 2**.

PARTITION TIER	ALBUMENTATIONS & PREPROCESSING POLICY	WHY WE CHOSE THIS DESIGN
Tier-1 (≥400) vs. Tier-2 (20–399)	Tier-1: heavy geometric/color jitter · Tier-2: subtle affine transforms	Prevents overfitting on head classes while preserving delicate scale patterns on scarce tail species
Validation & Test Splits	Deterministic 320×320 letterbox + training-only RGB normalisation	Ensures zero test-time distortion and handles the 56% portrait orientation shift in Indian mobile uploads

DINOv2 UMAP Projections Confirm Clade Structure & Zero Split Shift





What DINOv2 (`dinov2_vits14`) Shows

- **Zero Hidden Domain Shift**: An observer-grouped classifier trying to separate `train` from `val` and `test` scores **AUC = 0.542 and 0.507** (≈0.50 random guess).
- **Strong Taxonomic Clustering**: Cross-photographer nearest neighbours match the **same genus 12× above chance (37.5%)** and **same family at 64.6%**.
- **Why Soft-Dimming Beats Hard Masking**: UMAP confirms soft-dimming preserves boundary scale context without leaking habitat shortcuts.

DINOV2 & QUALITY AUDIT DIMENSION	MEASURED RESULT (31 OCT 2026 GPU RUN)	DIRECT MODELING DECISION
Train vs. Test Feature Overlap	Domain ROC-AUC = 0.507; median test-to-train species cosine = 0.569	Confirms post-2022 India test split is visually well-covered by the training distribution
Nocturnal Photo Preservation	Dropping dark photos deletes 13.6% of WHO Cat 2 & 67% of nocturnal species	Retain dark night photos (drop only blurry/tiny = 1.7%) and apply adaptive contrast correction

Disentangling Deadly Lookalikes & Unlisted Venomous Congeners

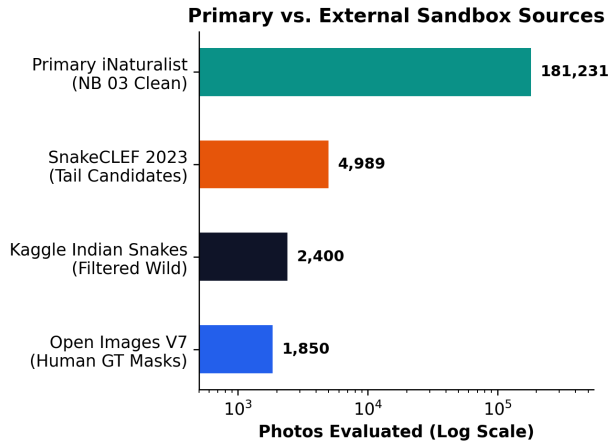


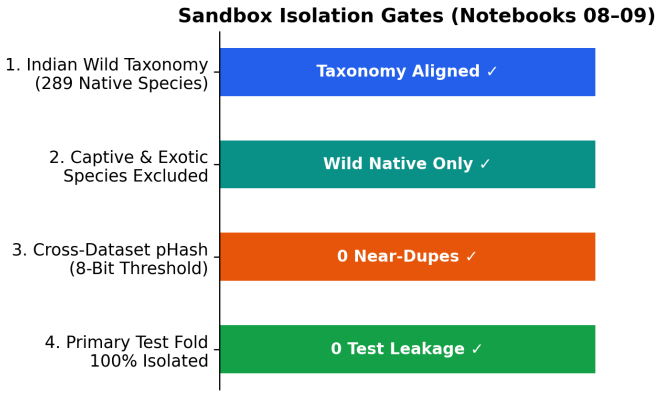
Preventing Fatal Misidentification

- **9 Benchmarked Lookalike Pairs**: Every pair has **≥70 test photos per side** (e.g., *Bungarus caeruleus* vs. *Lycodon aulicus / L. striatus*; *Daboia russelii* vs. *Python molurus*).
- **Sharpness Bias Mitigation**: Krait photos are 2.7× sharper than wolf snake photos ($p = 0.006$); blur jitter stops the model from cheating via image sharpness.
- **"WHO Not Listed" ≠ Safe**: DINOv2 shows top matches for WHO snakes are often **unlisted venomous relatives** (e.g., *Bungarus lividus*, 0.908).

CONFUSION PAIR TYPE	HIGH-RISK EXAMPLES IN DATASET	SAFETY MECHANISM IN SERPENTGUARD AI
Venomous vs. Harmless Lookalike	Common Krait vs. Wolf Snake (cos 0.842) · Russell's Viper vs. Rock Python (cos 0.782)	Asymmetric cost-sensitive loss heavily penalizing venomous → harmless false negatives
Venomous vs. Unlisted Congener	*Bungarus niger* → *B. lividus* (0.908) · *C. malabaricus* → *C. anamallensis* (0.902)	Family/Genus heads flag all Elapidae & Viperidae as venomous regardless of WHO list status

Augmenting Scarce Tail Species Without Contaminating Primary Splits





Strict Cross-Dataset Isolation

- **External Tail Candidates (NB 08)**: Harvests **SnakeCLEF 2023**, **Kaggle Indian Snakes**, and **Open Images V7** to boost scarce Tier-2 species.
- **Ground-Truth Mask Benchmark**: Uses human-annotated snake masks from Open Images V7 to benchmark SAM 2.1 segmentation IoU.
- **8-Bit pHash Sandbox Gate (NB 09)**: Filters out exotic/captive snakes and runs cross-dataset **pHash (≤8 bits) + ID checks** so zero test images leak into training.

SANDBOX DATASET	VERIFICATION RULE	ROLE IN PROJECT
`serpentguard-extended-sources`	Aligned to 289 native Indian species; captive/exotic species removed	Provides supplementary training examples for rare Tier-2 classes & Open Images V7 GT masks
`serpentguard-extended-combined`	Cross-dataset 8-bit pHash & observation ID deduplication against `val`/`test`	Isolated experimental sandbox ensuring zero contamination of our primary benchmark

From Verified Segmentation Dataset to Milestone 3 Model Training

MILESTONE 2 DELIVERABLE	KEY VERIFIED METRIC	MILESTONE 3 MODELING ACTION
1. Leak-Free Clean Dataset	181,231 photos · 208 classes · 0 observer/pHash overlap	Train multi-task backbones (MobileNetV3 / ConvNeXt-Tiny) mapped strictly via `classes.csv`
2. SAM 2.1 Soft-Dimmed Crops	129,241 masks · 4.2% hand cutouts · 80% bg dimming	Feed 320×320 letterboxed soft-dimmed tensors so classifiers learn scale patterns, not background
3. Long-Tail & Bias Controls	180:1+ skew · ≤1,000/class & ≤50/observer caps	Combine $1 / \sqrt{N_c}$ WHO safety sampler (1.6× Cat 1, 1.35× Cat 2) with logit-adjusted focal loss
4. Open-Set & Clade Safety	81 Unknown sp (656 photos) · 14 families · 77 genera	Jointly predict **Species + Genus + Family + WHO Risk** with calibrated open-set abstention

Group 5 Project Team — IIT Madras BS in Data Science & AI Lab (2026 T3)

JAYESH BANSAL · TEAM LEAD	VIKAS · TEAM MEMBER	PUNEET BAJAJ · TEAM MEMBER	MANAS RASTOGI · TEAM MEMBER
Project Architecture, Problem Formulation & Safety Evaluation	Taxonomic & WHO Clinical Risk Analysis, Lookalike & Literature Audit	iNaturalist Data Engineering, SAM 2.1 Mask Pipeline & Kaggle Datasets	Exploratory Data Analysis, DINOv2 UMAP Audit & Preprocessing Pipeline